



Pritam Dalal

Address: Gertrud-Grunow-Straße 32, 80807 Munich

E-Mail: pritamdadal@gmail.com

Mobile: +49 (0)176 8080 4720

LinkedIn: <https://www.linkedin.com/in/dalalpritam>

PROFILE SUMMARY

Test specialist with 8+ years of experience in automotive eMobility systems, specializing in component level validation, test automation and hardware-software integration. Strong background in HV power electronics, EV charging systems, inverters and HiL testing. Proven expertise in test automation, functional and non-functional testing, functional safety and field failure/root cause analysis.

EXPERIENCE

Jun 2019 - Present

Test and Validation Engineer | IAV GmbH, Munich

- Analyzed 500+ safety-relevant and functional requirements, creating test strategies and test specifications aligned with functional safety criteria
- Designed, implemented and maintained ECU-Test based automated test frameworks for integration, system and end-to-end testing
- Developed reusable test libraries and utilities to improve test coverage, reliability and execution efficiency, reducing total testing time up to 37%
- Designed and executed automated regression tests across changing requirement baselines, supporting continuous integration and release readiness
- Integrated automated test suites into Git-based workflows, enabling early defect detection and faster feedback cycles
- Collaborated closely with software developers, system engineers and hardware teams to define test strategies and acceptance criteria
- Performed in-depth hardware failure analysis on EV charging units and inverters, contributing to reliability and safety improvements
- Actively contributed to improving test processes, tooling and best practices across teams, bringing down release defects by 25% and improving delivery
- Analyzed test results and logs to identify flaky tests and improve pipeline stability
- Deployed multiple Python scripts to streamline and automate report generation, achieving 90% reduction in time
- Collaborated with international clients to develop ISO 26262 compliant automated testcases, improving functional safety coverage by 70%

May 2017 - Jun 2019

Development Engineer | IAV GmbH, Munich

- Conducted HiL testing for automotive embedded software in 12V and 48V vehicle systems
- Evaluated power system stability and failure modes for EV inverters under safety-critical conditions
- Validated system behavior using CAN bus data, signal analysis and ECU diagnostics
- Implemented Python scripts to compare HiL results with MiL simulations
- Supported testing of safety-relevant and real-time systems by defining robust test scenarios and edge cases

Sep 2016 - Mar 2017	Master Thesis IAV GmbH, Munich - Developed a software tool for complex CAN and FlexRay network analysis - Implemented data analysis using Python, Qt and MySQL
Jan 2016 - Aug 2016	Work Student Nippon Seiki, Munich - Created requirement-to-test traceability matrices using Python and XML test reports - Supported requirement analysis and documentation for OEM projects - Developed automation tools for internal reporting and data analysis
Nov 2012 - Sep 2014	Assistant Manager Tata Chemicals Limited, India - Led commissioning and maintenance of HV electrical systems - Applied Six Sigma methodology to investigate and determine root causes of induction motor failures, improving motor reliability by 30% - Managed multi-disciplinary technical teams
Aug 2010 - Nov 2012	Senior Electrical Engineer Vedanta Resources, India Commissioned motors & transformers along with electrical maintenance

EDUCATION

Oct 2014 - Mar 2017	Master of Engineering FH Rosenheim, Germany Electrical and IT (Specialization: Automation and Control)
Jul 2006 - May 2010	Bachelor of Technology NIT Allahabad, India Electrical Engineering

CORE SKILLS	
Standards	ISO 26262 (V-Model, verification and validation, traceability), Functional Safety Testing and Compliance Support, Failure Analysis and Root Cause Investigation
Embedded Software	ECU and Embedded Software Validation, CAN / LIN Bus Analysis, HiL / MiL Testing (dSPACE, MicroNova), EV Charging Units and Inverters
Testing / Automation	Python Test Automation, Regression and CI-based Testing, Requirement Traceability (DOORS, Codebeamer), Test Strategy and Test Specification

TOOLS

Softwares	ECU-TEST, CANalyzer, INCA
Programming	Python (NumPy, Pandas), SQL, VBA
Requirements	DOORS, Codebeamer
Agile tools	JIRA, OCTANE, ALM
Knowledge	OpenCV, Six-Sigma, PostgreSQL, Power BI
Languages	English - Fluent (C1), German - Professional working proficiency (B2)

CERTIFICATIONS

- ISTQB - Certified Tester Foundation Level
- Databases and SQL for Data Science with Python (Coursera)
- Machine Learning with Python (Coursera, IBM)
- Introduction to Deep Learning & Neural Network with Keras (Coursera, IBM)